

Estimation and Analysis of Consumer Demand for Intrastate Audio Incarcerated Peoples' Communication Services in Southern U.S. Jails



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"I am in great pain, and my hemorrhoids are more troublesome to me than the French Revolution was to the bourgeoisie."

-Karl Marx to Friedrich Engels, around 1865 [5]

Declaration

C.A. Kerr is currently serving as a data science intern in the Office of the Bureau Chief of the Enforcement Bureau at the Federal Communications Commission. The opinions expressed in this thesis do not reflect those of the bureau, commission, or any of its commissioners.

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Abstract

I aim to better understand the mechanisms in the United States (U.S.) incarcerated people's communication services market (IPCS), beginning with consumer demand. Understanding demand for audio IPCS from incarcerated people is important as it informs the auction mechanism in which IPCS contract terms are determined. I begin by utilizing data from the Department of Justice's (DOJ) 2019 Census of Jails to study differences in realized demand and pricing across jail types. Then, I utilize novel data gathered and compiled from the Federal Communications Commission's (FCC) Electronic Comment Filing System (ECFS) within a joint structural equation modeling (SEM) framework to estimate a consumer demand function for intrastate audio IPCS services in U.S. jails. Finally, I end with an examination of a variety of counterfactual regulation schemes and their welfare implications.

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Chapter 1

Preliminaries

1.1 Introduction

In the Federal Communication Commission's (FCC) 2024 Report & Order "In the Matter of Incarcerated Peoples' Communications Services (IPCS); Implementation of the Martha Wright-Reed Act and Rates for Interstate Inmate Calling Services", henceforth known as the *2024 Report & Order*, the IPCS market in the United States is characterized as one far away from those described in introductory microeconomics texts [3]. Instead, the IPCS market is described as a market full of failures resulting in a lack of competitive levers, constraining the setting of rates and fees that incarcerated people (consumers) must pay to communicate with those outside of their jail or prison. ([3]). Market failures are not uncommon in the United States and the global economy writ large, but the failures found in the US IPCS market have had deep and unique consequences for its marginalized and vulnerable consumers: incarcerated people.

To begin the FCC's July 18th, 2024 Open Meeting, Ulandis Forte, the grandson of the late Martha Wright-Reed, testified that his family's inability to afford collect calls from him while he was in prison left him feeling lonely, fearful, and robbed of many of life's great milestones

while he was incarcerated [3]. Simply, “she [Martha Wright-Reed] couldn’t afford to pick up”, just to hear that he was “alive and safe” [6]. For him, market failure in US IPCS did not simply mean he would have to incur an economic inconvenience, but a clear and longstanding detriment to both his and his family’s welfare.

Since at least 2013, the FCC has considered policies to realign the IPCS market into a fair and reasonable competition, and in July of 2024, the commission introduced a comprehensive set of reforms including rate caps and even out-right bans on certain types of service charges [2]. I do not attempt any kind of ad hoc policy analysis, but instead recognize that a policy’s effectiveness may well rely on the nature of the market mechanisms on which it is enacted and thus reexamine the same market during the same period that the commission did in their own policy research in hopes of drawing out these natures and their key properties [3].

This work also contributes to economics literature by the use of novel data and the study of an under-researched market. In fact, little to no published economic research exists on the IPCS market and none utilizing the data set used here. What literature does exist [4] has focused on the auction mechanism that determines IPCS contract terms. This is surely an important mechanism to draw implications from in determining the effectiveness of many policy counterfactuals, especially on the supply side of the IPCS market. However, to fully understand the auction mechanism and the IPCS market in full, the consumer demand function must also be characterized, as both bidders and auctioneers must take into account consumer demand when searching for the equilibrium contract terms. Additionally, while most policy counterfactuals considered in previous literature [4] and by the FCC [3] have revolved around directly regulating the prices of per-minute rates and ancillary service fees, an alternative approach to be considered by state and municipal agencies could be to restructure the auction mechanism (e.x. new scoring rules). A consumer demand function would

inform the consumer welfare implications that would be useful to estimate under this kind of policy counterfactual. Lastly, it should also be noted that while [4] solely focused on state prison systems, I solely focused on jails, a subset of IPCS that the FCC took a keen interest in during their past policy discussions [3]. Note that the FCC and DOJ both define jails as facilities that "typically hold inmates sentenced to jail facilities who usually have a sentence of 1 year or less" [3].

I begin by comparing demand elasticity and pricing for audio IPCS services across various jail types based on race, gender, conviction demographics, and jail conditions. I then move to more formally estimate consumer (prisoner) demand (including marginal utility and elasticity with time opportunity cost considerations) for phone calls and call minutes in U.S. jails using a joint structural equation modeling (SEM) procedure. Lastly, I use the resulting estimated parameters from the SEM procedure to simulate six different counterfactual policy scenarios on the regulation of intrastate per-min rates and ancillary service fees, with particular attention to the FCC's new policy of capping per-min rates according to an ADP-based schedule and banning fees. Each of these analyses is based on data from a single firm (City Tele Coin Company) in 2022.

The preliminary (comparative) empirical results for intrastate audio communications support a demand model that implies a demand for calls and call minutes that takes into account both ancillary service fees and per-minute rates. These empirical results suggest that per-minute rates impact demand for call minutes. The initial empirical results concerning demand elasticity also suggest that inmates in overcrowded jails and those in jails with a relatively high proportion of inmates under federal holdings may (individually) have stronger demands for longer calls than their counterparts and that the latter may also generally prefer to make more numerous calls relative to those in other facilities. The initial empirical results concerning

pricing suggest that inmates in jails with higher average daily populations (ADP) experience higher auto-pay fee frequencies and that jail populations with higher shares of Hispanic inmates and those who have higher demanded calls experience lower such frequencies. Lastly, these initial empirical results on pricing suggest that urban jails and jails with higher demanded call minutes, higher proportions of Black inmates, and higher proportions of felons (individually) all experience lower per-minute rates.

The SEM results produce an estimated mean taste shock for call minutes jointly with a standardized fixed utility (the fixed utility/disutility of making a call). These results then support a counterfactual analysis that suggests that fees noticeably improve revenue for the IPCS provider and that, by itself, a rate cap (whether ADP-based or uniform) does not greatly impact revenue while it greatly improves consumer welfare. Lastly, it was only with a rate cap and a fee ban counterfactual that the propensity to make a call rose above 0.50. Overall, these results show that rate-capping and fee-banning policies can substantially impact consumer welfare (through multiple behaviors) and, to a lesser extent, revenue, meaning regulators may be able to carefully choose policies to optimize on specific priorities such as increasing consumer welfare or the number of intrastate call made.

1.2 Background

When incarnated people want to communicate with those outside of their facility through audio or video communication, they must do so through services offered by an IPCS provider and at prices set by an exclusive contract between said provider and the relevant correctional authority. Some have argued that these prices are those of a monopoly, since the governing contract is exclusive, and the incarnated people are fully price-takers. The FCC illustrates this nicely in their *2024 Report & Order*, “Consumers have no means of switching to another

provider and no means of redress even if the IPCS provider ‘raises rates, imposes additional fees, adopts unreasonable terms and conditions for use of the service, or offers inferior service’” [3].

These contracts typically contain information on the quality of services, security measures, per-minute rates, and commissions. Of these, only per-minute rates are extensively considered here. The per-minute rates are almost always written as a uniform per-minute price for each call, though sometimes the first minute of a call has an increased price or connection fee that is incorporated into the total rate. Now, while I do not explore commissions in detail, understanding them is important for context in the IPCS market. Commissions are payments made from either the IPCS provider or the correctional authority to the other party, though almost always from the IPCS provider to the correctional authority. These commissions can be predetermined lump sums or be variable and determined by revenue or milestones. In either case, commissions can more or less be seen as legal kickbacks in exchange for the exclusive service rights the contract grants.

Across the United States, contract terms are determined by scoring auctions [4]. This means that in order to vie for a service contract, providers submit sealed bids to the correctional authority, i.e. proposed contract terms (rates, commissions, etc.), that are then ‘scored’ along a public rubric (included in the contract-seeking announcement). Once the bids are all submitted and scored, the provider with the best score wins the contract.

In appearance, the physical system in which incarcerated people use to make calls is similar to public pay telephones. However, in practice there are a few key differences. The first key difference is that the calls are paid for by either a collect or prepaid scheme. The collect call scheme asks the party outside of the correctional facility to accept responsibility for payment,

for which they will be sent a bill for at a later date. The prepaid scheme asks incarnated people to load funds onto a debit account that can then be used to pay for calls. The prepaid scheme is particularly interesting here, as incarnated people may incur a large variety of fees associated with this payment scheme, including but not limited to, 3rd party transaction fees, account setup fees, automated payment fees, and single-call fees (off-account). Though the collect call scheme is not without its own fees, including but not limited to, paper bill/statement fees, live agent fees, and interactive voice response fees.

Chapter 2

Descriptive Data Analysis

2.1 Data Description

I source data from two main government agencies. The Federal Communications Commission via their Electronic Comment Filing System and the Department of Justice (DOJ) via their 2019 Census of Jails.

2.1.1 FCC ECFS Submissions

Since 2014, the Federal Communications Commission has required that service providers in the IPCS market submit a number of forms related to their per-minute rates for phone calls, ancillary service fees, commissions, operating costs, and more. I utilize two of these forms: FCC Form 2301 and FCC Form 2303. The official submissions of these forms are under a protected order, meaning that they are only available to certain FCC employees and outside counsel for the service providers or inmates. However, public redacted versions of these submissions were available via the FCC ECFS. All of the data provided within these forms is

submitted and redacted (or not redacted) by a representative of the service provider.

FCC Form 2301

FCC Form 2301 is known as the annual reporting form for IPCS providers and primarily contains information on the set rates, commissions, and ancillary service fees for each facility that contracted with a provider in the relevant year. The form has been in use since 2016, though I only consider submissions since 2021.

FCC Form 2301 asks providers to submit first minute and additional minute per-minute rates for pre-paid and collect call intrastate and interstate audio communications for each facility in each year. Using this data, I calculate the 15-minute intrastate and interstate rates ($\text{first-minute} + 14 * (\text{additional-minute})$). I chose to use the 15-minute rate, because its costs are a bit easier to visualize than those of the average rate and because the average rate is much more difficult to estimate (I would need to know the expected length of a call for each rate). Additionally, I calculate the fees set for each facility during each year. In most cases, the fees were given as a set dollar amount, and so no calculation was needed. However, in some cases the fees were reported as an interval of prices or as a percentage of the other costs of the call. In the first case, the average of the two endpoints of the interval was taken and used as the set fee going forward. In the latter case, fees were calculated based on a percentage of a 15-minute phone call at the given rate. Within FCC Form 2301, commissions were very rarely left unredacted and thus were not usable. The providers also submitted the average daily population (ADP) of each facility during each year, but as discussed later, these data points are often unreliable.

FCC Form 2303

FCC Form 2303 contains information primarily related to expenses and operating costs for the service provider specifically in 2022. I do not utilize the majority of this information and instead focus on the providers' reports of facility-specific demand for IPCS services, including both calls and fees for ancillary services. For FCC Form 2303, providers reported the total number of audio communications (phone calls) and the total number of minutes serviced during these communications with respects to call location (interstate or intrastate) and facility. For the fee-related submissions, the providers submitted facility-specific revenues and counts of how many times a fee was charged for four different types of fees: automated payment, live agent, billing, and third-party transaction. The providers also submitted the average daily population (ADP) of each facility during 2022, but as alluded to earlier and discussed later, these data points were often unreliable. Despite FCC ECFS containing 15 submissions of this form only one provider's (City Tele Coin Company) submission was usable. After cleaning, 190 complete or nearly-complete observations became available for facilities that contracted with City Tele Coin Company in 2022 (2 had some missing information on ancillary service fees).

2.1.2 DOJ Census of Jails

In addition to data from the FCC ECFS, I leverage some information from the Department of Justice's 2019 Census of Jails (all of the data points are actually in fact related to June of 2019). This census report provides information on city, county, state, and federal facilities (2,924), all of which, except for federal facilities, are classified as jails. Recall that, both the FCC and DOJ classify jails as facilities that "typically hold inmates sentenced to jail facilities who usually have a sentence of 1 year or less", or in layman's terms, short-term correctional facilities [3]. The census contains multitudes of data points on jail characteristics

from demographics to physical conditions. I pull data facility-specific points on the shares of females and juveniles in addition to indicators on whether or not the jail is overcrowded, privately operated, and in an urban area. I also pull jail-level (if a jail-system has multiple facilities) data points on the shares of African-Americans, Hispanic people, U.S. citizens, confined inmates (as opposed to paroled), convicted inmates, inmates held under felonies, and inmates held for federal authorities. The DOJ Census of Jails also reports an ADP metric, which I use instead of the ADP metric provided by the IPCS service providers in FCC Form 2303, simply because unlike the ADP metric from the providers, it is available for every facility. More information on these numbers can be found in Appendix A.

2.1.3 Merging

After extensive cleaning, I merge both the DOJ Census of Jails and FCC Forms 2301/2303 data sets into a singular data table. An issue that arises during this process was that the DOJ and FCC defined facilities slightly differently. The DOJ uses the physical facility site as the facility even if there are separated halls, buildings, etc., while the FCC separates by jurisdiction. For example, if both a county and a state correctional authority share a physical location, then the service provider submitting to the FCC will report these as two facilities while the DOJ will report this site as one facility. For most variables, I simply aggregate up to the DOJ-level via summation. The interesting items here are the per-min rates. Perhaps the reason that the FCC asks for these kinds of facilities to be separated is because IPCS service providers can and do charge different rates to different sections of the facilities. Luckily, since in each of these cases, the 2022 ADP (ADP^{2022}) from FCC Form 2301/2303 was available, to aggregate the rate to the DOJ level I am able take a weighted average of the per-min rates faced by the prisoners their respective facilities, where the weight for section j of facility i with k sections is,

$$w_{ij} = \frac{ADP_{ij}^{2022}}{\sum_{j=1}^k ADP_{ij}^{2022}} \quad (2.1)$$

After this calculation, I solely utilize the 2019 ADP (*ADP*) as reported in the DOJ Census of Jails. Lastly, note that this weighting scheme does make the assumption that the relative distribution of prisoners across sections of a facility was similar between 2019 and 2022. A detailed description of this merged data set can be found in Appendix A.

2.1.4 City Tele Coin Company Subset

For the rest of the paper, I will be concerned with a single firm in a single year, City Tele Coin Company (CTC) in 2022. CTC is a mid-size regional IPCS provider that services jails primarily in the southern United States (Mississippi, Arkansas, Kansas, Texas, Missouri, Louisiana, and Oklahoma). I chose to subset the analysis to this single year and firm primarily due to data constraints. First, CTC was the only firm in which the expected cost from fees could be derived, and secondly, because it was the only firm with complete realized demand (number of intrastate calls and billed minutes). Table 2.1 below compares characteristics related to CTC and the jails it served in 2022 to that of the rest of the firms observed in FCC Form 2301, FCC Form 2303, and the 2019 Census of Jails [DOJ].

From Table 2.1, it appears that, generally speaking, the subset of the data concerning CTC in 2022 is quite representative of the larger sample, with two notable exceptions. First, the 15-min intrastate rates are much higher in the CTC subset than the larger sample, and the distribution of these rates are a good bit tighter in the sample. The distribution of the CTC subset is of no concern at all, as its relative tightness ought to be expected of a subset with a shared source (CTC). The first does qualify the subset as having higher than average 15-min

Variable	CTC Median	CTC σ	2022 Median	2022 σ
ADP*	140.34	187.2689	107	22366.66
Intrastate 15-Min Rate*	5.25	0.826402	3	1.750633
Urban Indicator	0	0.27037	0	0.47681
Private Indicator	0	0.10233	0	0.09026
Overcrowding Indicator	0	0.49731	0	0.48426
Share of Females	0.1589	0.12908	0.1667	0.11544
Share of Juveniles	0	0.00885	0	0.01289
Share of Confined Inmates	1	0.02789	1	0.07399
Share of Convicted Inmates	0.3265	0.28034	0.3037	0.26283
Share of Inmates Who are U.S. Citizens	1	0.07684	1	0.10819
Share of Black Inmates	0.19022	0.24752	0.17413	0.22753
Share of Hispanic Inmates	0.1515	0.19781	0.03061	0.17930
Share of Felons	0.7792	0.23526	0.7048	0.25502
Share of Inmates Under Federal Holdings	0	0.11544	0	0.14230

Table 2.1 *CTC's 2022 Data Compared to that of the Larger Sample*. Note that the median is used instead of the mean. Especially in the larger sample, outliers play a large role in skewing some of the data. σ denotes the standard deviation. * indicates the data is sourced from FCC Form 2301 and 2303, while its absence indicates that the data is sourced from the 2019 DOJ Census of Jails [DOJ].

intrastate rates. However, the larger sample as a whole does have a larger enough spread ($\sigma \approx 1.75$) that the subset's mean is well within a reasonable bound, i.e. not extreme. The second item to note here is that the jails CTC serviced in 2022 appear to have a much higher than average proportion of Hispanic inmates (approx. their median 5-times the sample median). This finding is surely interesting, but its impact on the analysis remains to be seen.

2.1.5 Expected Costs from Ancillary Service Fees

In Chapter 2, I introduce expected cost from ancillary service fees as a fixed cost in our demand model and in Chapter 3 I utilize data from City Tele Coin Company (CTC) in 2022 (FCC Form 2301) to estimate such a demand model. This requires us to calculate the expected cost from ancillary service fees, which is a non-trivial calculation. So I take a brief

detour here to explain it.

In 2022, CTC charged uniform fees with respect to the type of fee across all of the jails to which it provided service. However, these fees were not charged at the same frequencies across all jails (2.2). The four ancillary services that CTC charged fees for in 2022 were auto pay, live agent, paper billing, and 3rd part transactions. I do not assume that these fees were charged independently of each other, and in fact, I allow for the possibility that the fees were charged conditionally on each other. To utilize the linearity of conditional expectation, I do not need independence of our random variables, which in this case are the charge frequencies of each fee, but I do need a unifying condition for our expectation. For this, I advantage of the structure of the data and utilize the fact that each observation of fee charge frequency is contained within a unique jail. In fact, each jail corresponds with a disjoint subset of fee charge observations (Two separate parties related to two separate jails are not both charged when an incarcerated person calls another incarcerated person) [3]. Thus, I can use the unique jail as a unifying condition for our charge frequency observations and subsequently calculate the per-call expected cost from ancillary service fees for each unique jail i as the sum of expected per-call costs from each fee type j using the linearity of conditional expectation,

$$\mathbb{E}[\mathbf{F}_i|J_i] = \sum_{j=1}^4 C_j \mathbb{P}[f_j|J_i] \quad (2.2)$$

where C_j is the constant cost of fee type j , f_j is an indicator on whether or not fee type j is charged, and $\mathbf{F}_i$ is the total per-call cost from ancillary service fees in jail i .

2.2 Summary Statistics

We begin the empirical sections of our analysis with a specific treatment of data related to a single firm in 2022: City Tele Coin Company. This is the only firm in which we were provided with usable realized demand data, as described in Chapter 1. In order to summarize to basics of the data and setting, we illustrate two key relationships: how set rates and fees vary by facility characteristics and demand, and how demand varies by set rates and fees. A detailed summary of the variables used in this section can be seen in Appendix A.

2.2.1 Rates and Fees

As previously mentioned, both the interstate and intrastate per-minute rates are present in our data. However, we focus solely on intrastate rates here as interstate are regulated differently than intrastate rates (in that intrastate rates are more or less unregulated in our data). The Federal Communications Commission has and has had for many years a clear jurisdiction on interstate calls, one that has not been enjoyed for intrastate calls [3]. This result is illustrated within the distributions of interstate and intrastate 15-minute call rates in Figure 2.1.

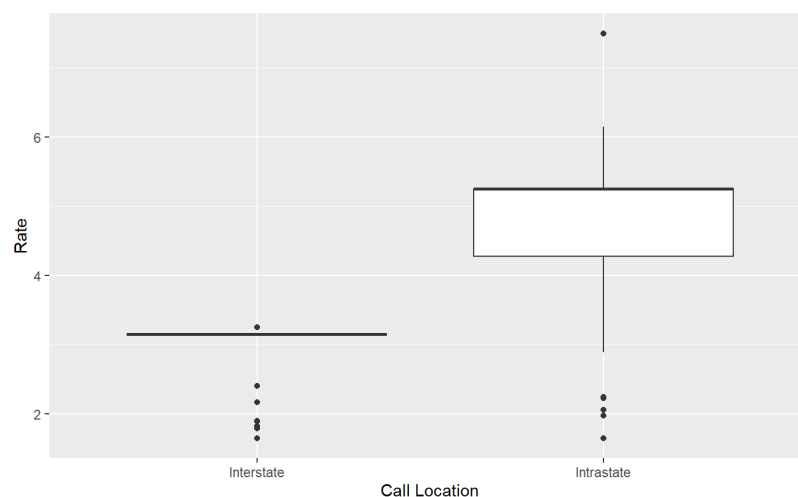


Fig. 2.1 *15-Minute Call Rate Distribution by Call Location* Note that there appears to a be a clear and obvious cap for interstate rates, while in juxtaposition intrastate rates seem to enjoy much more variation.

It is also true that interstate calls do not interact with ancillary service fees in the same way as intrastate calls, but this can be quite complicated [3].

In order to study fees, we calculated the frequency in which each type of fee was charged per phone call. This metric is especially useful because City Tele Coin Company charged all of their facilities the same fees for 2022. The distributions of these charge frequencies are illustrated in Figure 2.2.

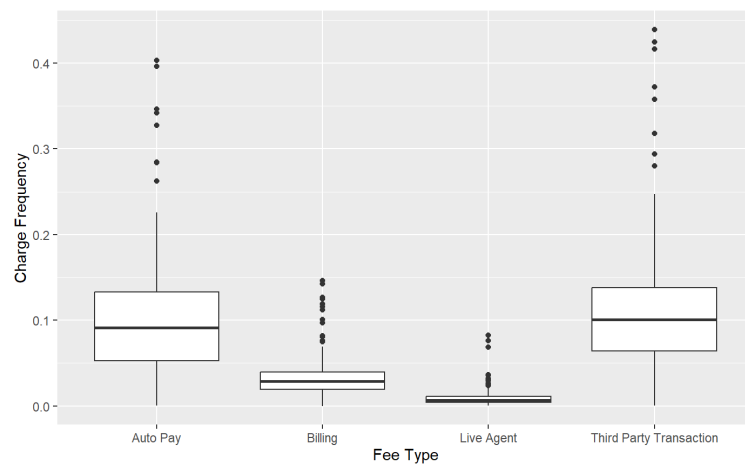


Fig. 2.2 *Charge Frequency by Fee Type* Notice that the two fees with the most variation by far are auto pay and third-party transaction.

There is some variation in the absolute fee from a third-party transaction in this data set, as this kind of fee is often a percentage of the rest of the cost of the call. For the sake of simplicity, we focus on auto pay fees for the remainder of this section.

The below summary (Table 2.2) is for an OLS model on the 15-minute intrastate rates faced by each facility.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.8236	2.3235	4.23	0.0000
ln(Total Intrastate Minutes Serviced)	-0.1512	0.0502	-3.01	0.0030
Overcrowding Indicator	-0.0539	0.1187	-0.45	0.6503
Private Indicator	0.4895	0.5602	0.87	0.3835
Urban Indicator	-0.4844	0.2318	-2.09	0.0381
Share of Female Inmates	0.5957	0.4773	1.25	0.2138
Share of Juvenile Inmates	-11.9466	6.6405	-1.80	0.0738
Share of Confined Inmates	-2.1573	2.0997	-1.03	0.3057
Share of Black Inmates	-0.7925	0.2855	-2.78	0.0061
Share of Hispanic Inmates	-0.0993	0.3543	-0.28	0.7796
Share of Inmates Who are U.S. Citizens	-0.5145	0.7851	-0.66	0.5132
Share of Convicted Inmates	-0.4086	0.2145	-1.90	0.0585
Share of Felons	-0.5698	0.2640	-2.16	0.0323
Share of Inmates under Federal Holdings	0.1199	0.5466	0.22	0.8267
Share of Black Inmates * Share of Hispanic Inmates	-1.5406	2.5097	-0.61	0.5401

Table 2.2 *OLS Model on 15-minute Intrastate Rates Faced by Facility*

The key takeaways from the above output are that facilities with higher demand faced lower rates, urban jails experienced lower rates than non-urban jails, and that rates decrease with the shares of Black inmates and felons. An analysis of deviance (ANOVA), resulted in a p-value of 7.3×10^{-5} , which suggests a good fit.

The next and last summary (Table 2.3) describes an OLS model, just as the one before, but this time, on auto pay fee charge frequencies instead of intrastate rates.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.7821	2.2586	-2.56	0.0114
ln(Total Intrastate Calls)	-0.2979	0.0733	-4.06	0.0001
Overcrowding Indicator	-0.0512	0.1156	-0.44	0.6583
Private Indicator	0.2797	0.5404	0.52	0.6055
Urban Indicator	-0.1726	0.2346	-0.74	0.4629
Share of Female Inmates	0.6103	0.4597	1.33	0.1861
Share of Juvenile Inmates	0.4483	6.4363	0.07	0.9446
Share of Confined Inmates	3.2779	2.0251	1.62	0.1074
Share of Black Inmates	0.1018	0.2733	0.37	0.7099
Share of Hispanic Inmates	-0.7630	0.3446	-2.21	0.0282
Share of Inmates who are U.S. Citizens	1.1115	0.7582	1.47	0.1446
Share of Convicted Inmates	-0.0094	0.2085	-0.04	0.9642
Share of Felons	-0.0748	0.2535	-0.29	0.7684
Share of Inmates Under Federal Holdings	-0.7540	0.5232	-1.44	0.1514
ln(ADP)	0.4578	0.0753	6.08	0.0000
Share of Black Inmates * Share of Hispanic Inmates	1.8548	2.4356	0.76	0.4474

Table 2.3 *OLS Model on Auto Pay Fee Frequency Faced by Facility*

The key takeaways from the above output are that facilities with higher demand faced lower auto pay fee charge frequencies, the auto pay fee charge frequencies increased with average daily population, and that auto pay fee charge frequencies decreased with the shares of Hispanic inmates. An analysis of deviance (ANOVA), resulted in a p-value of 2.6×10^{-3} , which suggests a good fit.

2.2.2 Demand Elasticity

We begin to describe the demand elasticity from both per-minute rates and fees with the simple illustration in Figure 2.3.

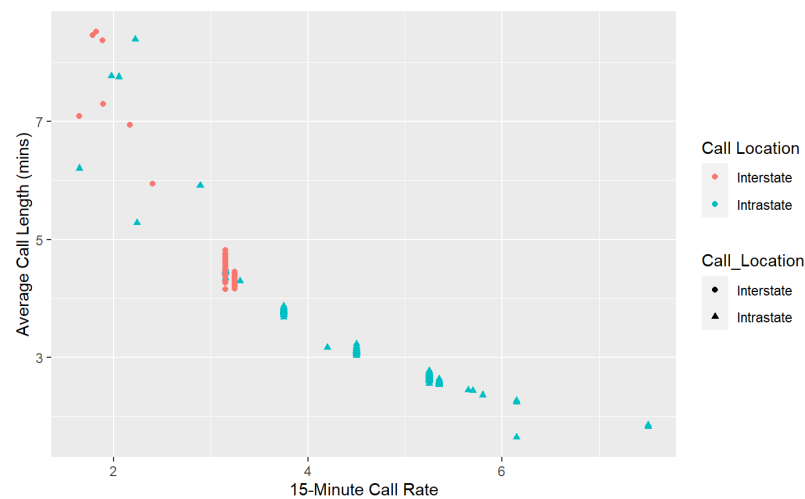


Fig. 2.3 *Average Call Length by 15-minute Call Rate* Notice that, at least for intrastate calls, call lengths appear to decrease exponentially with the 15-minute call rate. This matches what we assumed in our theoretical demand function: that call length is a Poisson random variable.

The below summary (Table 2.4) is for a quasipoisson model on average daily calls per inmate made in each facility.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.8915	2.3435	-1.66	0.0987
ln(weighted Intrastate 15-min Rate)	0.3343	0.2363	1.41	0.1591
ln(Auto Pay Fee Frequency)	-0.3253	0.0771	-4.22	0.0000
Overcrowding Indicator	0.0458	0.1320	0.35	0.7288
Private Indicator	0.7306	0.2246	3.25	0.0014
Urban Indicator	-0.7128	0.1805	-3.95	0.0001
Share of Females	0.3862	0.6133	0.63	0.5297
Share of Juveniles	-2.4838	7.1700	-0.35	0.7295
Share of Confined Inmates	1.1571	2.0058	0.58	0.5648
Share of Black Inmates	0.0301	0.3502	0.09	0.9317
Share of Hispanic Inmates	-0.9893	0.6571	-1.51	0.1341
Share of Inmates who are U.S. Citizens	0.6213	0.9195	0.68	0.5002
Share of Convicted Inmates	0.1313	0.2447	0.54	0.5923
Share of Felons	0.8956	0.3643	2.46	0.0150
Share of Inmates under Federal Holdings	0.9674	0.2679	3.61	0.0004
Share of Black Inmates * Share of Hispanic Inmates	3.6421	4.2662	0.85	0.3945

Table 2.4 *Quasipoisson Model on Average Daily Calls per Inmate by Facility*

The key takeaways from the above output are that auto pay fee frequencies are lower demand for phone calls (fixed cost increase), demand for phone calls is lower among urban jails than non-urban jails, demand for phone calls is higher among private jails than public jails, and that demand seems to increase with the share of felons and convicted inmates (in jail longer). The quasipoisson model was chosen because choosing how many phone calls to make each day is logically a Poisson process. An analysis of deviance (ANOVA), resulted in

a p-value of approximately zero, which suggests a good fit. Lastly, the dispersion parameter was calculated to be 22687.71, which suggests that quasipoisson was a good choice of distribution (accounts for overdispersion).

Moving forward, Table 2.5 shows a rate model (still quasipoisson), but this time for the total minutes serviced by facility over total audio communications serviced by facility (this is a rational number, which is a key assumption of a rate model).

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-8.4038	4.2330	-1.99	0.0487
ln(weighted Intrastate 15-min Rate)	0.5673	0.5134	1.11	0.2707
ln(Auto Pay Charge Frequency)	0.0436	0.1272	0.34	0.7324
Overcrowding Indicator	-0.0260	0.2169	-0.12	0.9048
Private Indicator	-1.7632	0.9136	-1.93	0.0553
Urban Indicator	-0.6689	0.3987	-1.68	0.0953
Share of Females	-0.3927	1.0531	-0.37	0.7097
Share of Juveniles	-2.7121	11.9262	-0.23	0.8204
Share of Confined Inmates	-0.1940	3.6114	-0.05	0.9572
Share of Black Inmates	-0.9318	0.5779	-1.61	0.1087
Share of Hispanic Inmates	0.1103	0.6791	0.16	0.8712
Share of Inmates who are U.S. Citizens	0.9393	1.5807	0.59	0.5532
Share of Convicted Inmates	-0.5540	0.4286	-1.29	0.1979
Share of Felons	-0.5408	0.5132	-1.05	0.2935
Share of Inmates under Federal Holdings	-2.8866	1.0457	-2.76	0.0064
Share of Black Inmates * Share of Hispanic Inmates	7.6819	5.7026	1.35	0.1798

Table 2.5 *Quasipoisson Rate Model on Average Length of Calls by Facility*

The key takeaway from this output is that The key takeaway from the above output is that it suffers from issues related to multicollinearity. As seen in Table 2.2, the share of

Black inmates is especially well correlated with lower faced rates. In fact, we calculate it's generalized variance inflation factor (GVIF) in the above model as 2.94 and the GVIF of its interaction term with the share of Hispanic inmates as 2.83. Both of these GVIFs are well above those of the other variables. As such, it is difficult to identify the effect of either on the realized demand for call minutes. Since we are generally focused on the effects from pricing policies (i.e. rates and fees), we remove the share of Black inmates as an effect and obtain much more reasonable results. This adjusted model is described in Table 2.6 below.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.7284	0.1037	26.30	0.0000
ln(Weighted Intrastate 15-min Rate)	-1.0039	0.0228	-44.07	0.0000
ln(Auto Pay Charge Frequency)	0.0418	0.0100	4.17	0.0000
Overcrowding Indicator	0.0571	0.0153	3.74	0.0003
Private Indicator	-0.0008	0.0241	-0.03	0.9722
Urban Indicator	-0.0149	0.0204	-0.73	0.4675
Share of Female Inmates	-0.1711	0.0739	-2.32	0.0217
Share of Juvenile Inmates	-0.7033	0.7605	-0.92	0.3564
Share of Inmates who are U.S. Citizens	0.0045	0.0967	0.05	0.9626
Share of Convicted Inmates	-0.0419	0.0285	-1.47	0.1439
Share of Felons	0.0197	0.0400	0.49	0.6231
Share of Inmates under Federal Holdings	0.2445	0.0247	9.90	0.0000
Share of Hispanic Inmates	0.0998	0.0562	1.78	0.0773

Table 2.6 *Quasipoisson Rate Model on Average Length of Calls by Facility without Share of Black Inmates*

Now we can interpret this kind of model. The key takeaways from the above output are that per-minute rates seem to lower demand for phone call minutes while auto pay fees have a relatively small effect on demand for call minutes, demand for call minutes is higher among overcrowded jails and jails with a relatively high number of inmates held under federal

authority, jails than non-urban jails, and that demand for call minutes seems to be lower among jails with a relatively large share of female inmates. The quasipoisson model was chosen because choosing how many minutes to stay on the phone per call is logically a Poisson process. An analysis of deviance (ANOVA), resulted in a p-value of approximately zero, which suggests a good fit. Lastly, the dispersion parameter was calculated to be 1055.614, which suggests that quasipoisson was a good choice of distribution (accounts for overdispersion).

Chapter 3

Structural Data Analysis

3.1 Theory

3.1.1 Motivations

We begin with a treatment of the consumer, incarcerated people, and their choices, in order to estimate a demand function for call minutes from the telephone providers. Later on, we will show how this demand function informs the decisions that service providers and correctional authorities make when setting contract terms (specifically per-minute rates and commissions). Lastly, this demand function will be important to consider when drawing implications from any policy counterfactual.

The following example illustrates the decision-making process faced by the incarcerated person,

David is a person incarcerated at some arbitrary county jail in the United States. For every week during his time served, he would like to call each member of his countably infinite family for a countably infinite number of minutes.

Additionally, David experiences some random and self-perceived 'need' to make a phone call (unobserved by the econometrician). For example, David's mother might fall ill one week, increasing his self-perceived 'need' to call home. Very importantly, these 'needs' are independent of each other and realized call minutes week-to-week, i.e. this week's 'need' is not affected by whether or not he called home last week or for how long. Now, given this week's 'need', David must choose how many calls to make and how long each call will last. David must consider two types of costs when making these decisions: fixed and variable. The only fixed costs for David to consider are some prison-specific ancillary service fees and a fixed endogenous cost (e.x. time to walk to phone booth). The total amount of fees is not necessarily constant across each call David may make, but they are constant across lengths of a call. In other words, the expected amount of ancillary fees David pays is the same for a 1-minute and a 20-minute phone call. While David only faces two fixed costs, he experiences twice the variable costs: one from a per-minute rate and another from a time opportunity cost. The per-minute rate is as prescribed in the contract between the service provider and correctional authority, and both fees are constant across calls, but clearly, the total variable cost is time-dependent.

3.1.2 Consumer Utility from Call Minutes

Given the process as described above, we formulate the following utility function for the incarcerated person (consumer) choosing the length of a single call (Equation 3.1). Note that a zero-minute call would imply that they chose not to make that call. Thus, we write the utility function as

$$u_{it}(l_{it}) = \mathbb{1}_{l_{it} > 0} [\beta_{it} * \ln(l_{it}) - l_{it} * r_{it} - (\mathbf{F}_{it} + \varepsilon_{it})] \quad (3.1)$$

Note that without the indicator, the utility function (2.1) would give that $u(0) = \infty$, which is certainly unreasonable. So we say that the utility is zero when $l_{it} = 0$.

Here in Equation 3.1, $u_{it}(l_{it})$ is the consumer's utility from making call i at week t for length $l_{it} \geq 0$. β_{it} is a coefficient that describes the consumer's per-minute benefit of the call. r_{it} is the per-minute, contractually prescribed rate of call i at week t . c_t is the time opportunity cost for the consumer at week t . Lastly, $\mathbf{F}_{it}$ is the total amount of ancillary fees the consumer pays for making call i at time t , and ε_{it} is the endogenous fixed cost of making call i at time t .

To be explicit, $\mathbf{F}_{it}$ is the sum of some subset of 4 individual fees the consumer may face: an automated payment service fee, a live agent service fee, a paper billing fee, and a 3rd party transaction fee.

Further, see that the consumer's marginal benefit from making an l -lengthed (for $l_{it} > 0$) call is,

$$\frac{\partial u}{\partial l_{it}} = \frac{\beta_{it}}{l_{it}} - r_{it} \quad (3.2)$$

Thus, the optimal length of call i and week t is,

$$l_{it}^* = \frac{\beta_{it}}{r_{it}} \quad (3.3)$$

Also, see that, as described in the example, F_{it} does not affect the consumer's marginal decision in regards to the length of the call.

We can now solve for the consumer's indirect utility function,

$$v_{it}^*(l_{it}) = \mathbb{1}_{l_{it} > 0} [\beta_{it} [\ln(\frac{\beta_{it}}{r_{it}}) - 1] - (\mathbf{F}_{it} + \varepsilon_{it})] \quad (3.4)$$

3.1.3 Propensity to Make a Call

While the previous section discussed the utility of each minute on a call for consumers, there is still a significant missing piece to the puzzle for consumer demand for IPCS audio services: the propensity of a consumer to make a call in the first place. We continue by describing this propensity as a probabilistic extension of the utility function (2.1) in the previous section .

To begin, we should make a few assumptions clear. First, we assume that conditional on a consumer making a call, they will choose to stay on the phone for the optimal number of minutes as described by 2.5. Secondly, we assume that, for each potential phone call, the consumer is aware of their own tastes (including benefits) and the per-minute rate that they will be charged but unaware (at least fully) of the exact costs from fees.

Then, under those assumptions, we say that a consumer will make a call if and only if they expect their utility from said call to be greater than the fixed cost of the call (costs from fees). In other words,

$$\mathbb{E}[v^*(l_{it})] = \mathbb{P}(l_{it} > 0 | r_{it}) \times [\beta_{it} [\ln(\frac{\beta_{it}}{r_{it}}) - 1] - (\mathbb{E}[\mathbf{F}_{it}] + \varepsilon_{it})] > 0 \quad (3.5)$$

Note that the variable component of 2.7 drops the expectation, because the consumer is fully aware of the values of β_{it} , r_{it} , and ε_{it} before making a phone call. Thus, we describe the propensity of a consumer to make a call as

$$\mathbb{P}(l_{it} > 0 | r_{it}) = \mathbb{P}\{\beta_{it} \in \mathbb{R} : \mathbb{E}[v_{it}^*(l_{it})] > 0\} \quad (3.6)$$

This will be a considerably important metric that is monitored in the upcoming counterfactual simulations in Chapter 5, along with call length, utility, and cost (as revenue).

3.2 Demand Function Estimation

Recall Equation 2.5 and our discussion in 2.3, namely Equation 2.7. In our data from City Tele Coin Company (henceforth CTC) in 2022, we observe $\hat{l}_t^* \approx l_t^*$ in the data, where $l_t^* = (l_{t_1}^*, \dots, l_{t_n}^*)$. In other words, we observe the optimal call lengths (in minutes) that consumers choose when making phone calls.

With that, we move to suppose that our β_{it} are exponentially distributed with parameter λ_t , and that our ε_{it} are normally distributed with mean μ_t and variance σ_t^2 . Additionally, we calculate $\hat{\beta}_{it} = r_{it} \hat{l}_{it}^*$ and assume that $\hat{\beta}_{it} \xrightarrow{d} \beta_{it}$. With these assumptions and related calculations, we can elegantly estimate our consumer demand function via maximum likelihood estimation.

To begin formulating our likelihood function, suppose there is a one-to-one measurable function $X : l_t^* \rightarrow \theta_t$ such that $X^{-1}(\theta_t) = l_t^*$, with θ_t being a vector-valued parameter containing λ_t , μ_t , and σ_t^2 .

Then the likelihood function $\mathcal{L}(\theta_t | l_t^*; r_t)$, can be expressed as

$$\mathcal{L}(\theta_t | l_t^*; r_t) = \mathbb{P}(l_t^* | \theta_t; r_t) = \prod_{i=1}^n \mathbb{P}([\hat{\beta}_{it} [\ln(\frac{\hat{\beta}_{it}}{r_{it}}) - 1] - \mathbb{E}[\mathbf{F}_{it}] > \varepsilon_{it} | \theta_t; r_{it}) \times \mathbb{P}(\hat{\beta}_{it} | \theta_t; r_{it}) \quad (3.7)$$

where $\arg \max \mathcal{L}(\theta | l_t^*; r_t) \xrightarrow{P} X$ and $n \in \mathbb{N}$.

For computational efficiency, we use the log-likelihood function for estimation.

$$\ln \mathcal{L}(\theta_t | l_t^*; r_t) = \sum_{i=1}^n \ln \mathbb{P}([\hat{\beta}_{it} [\ln(\frac{\hat{\beta}_{it}}{r_{it}}) - 1] - \mathbb{E}[\mathbf{F}_{it}] > \varepsilon_{it} | \theta_t; r_{it}) + \ln \mathbb{P}(\hat{\beta}_{it} | \theta_t; r_{it}) \quad (3.8)$$

3.3 Results

3.3.1 Estimates for Consumer Demand Function

As described in Chapter 3, I aim to estimate λ_t , μ_t , and σ_t^2 via a joint maximum likelihood procedure. In practice, I calculate the joint likelihood for a large number of combinations of these parameters, locate maxima and minima, and find which results in the highest likelihood. Using this technique, I was not able to identify either μ_t nor σ_t^2 . The explanation for μ_t comes simply from the structure of the model. ε_t is a linear shifter of demand and as such is inseparable from the utility portion of the demand function. Nonetheless, λ_t is identifiable. Figure 3.1 shows the maxima in which the best estimate of λ_t is found. Since μ_t and σ_t^2 were not identifiable, I fix them at 0 and 1, respectively.

This leads us to the following estimate of the consumer utility,

$$u_{it}(l_{it}) = \mathbb{1}_{l_{it} > 0} [(0.935) * \ln(l_{it}) - l_{it} * r_{it} - \mathbf{F}_{it}] \quad (3.9)$$

Note that I fit β_{it} and ε_{it} with their respective estimated or fixed means.

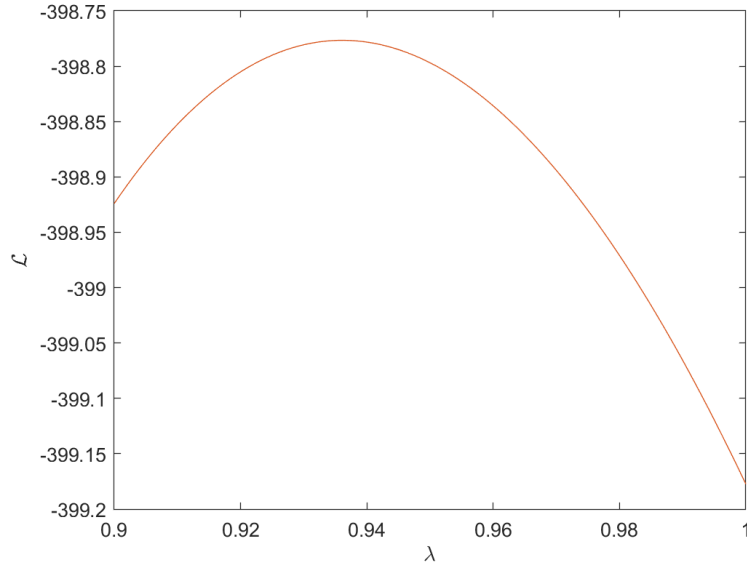


Fig. 3.1 *Joint Likelihood Estimation of λ_t* λ_t appears to be estimated at 0.935. Note that this estimation holds both in the joint and univariate sense.

I also formulate a fit for the expected indirect utility.

$$\mathbb{E}[v^*(l_{it})] = \mathbb{P}(l_{it} > 0 | r_{it}) \times (0.935) \left[\ln\left(\frac{0.935}{r_{it}}\right) - 1 \right] - (\mathbb{E}[\mathbf{F}_{it}]) \quad (3.10)$$

Equation 4.2 will be extensively used in the next section on welfare. Before moving into that analysis though, note that Equation 4.2 does imply that

$$\mathbb{P}(l_{it} > 0 | r_{it}) = \mathbb{P}\{\beta_{it} \sim \exp(0.935) \in \mathbb{R} : [\beta_{it} \left[\ln\left(\frac{\beta_{it}}{r_{it}}\right) - 1 \right] - \mathbb{E}[\mathbf{F}_{it}]] > \varepsilon_{it} \sim \mathcal{N}(0, 1)\} \quad (3.11)$$

which is important to the welfare calculations that I will work through from here on out.

3.3.2 Welfare Analysis

Consider p_{it} to be an incarcerated person i in period t . The incarcerated person can be characterized as being of type T_k by their β_{it} and ε_{it} . Thus, let A be the disjoint union of

all A_k that collectively contain all β_{it} and F be the disjoint union of all F_k that collectively contain all ε_{it} , such that

$$p_{it} \in T_k \iff \{\beta_{it}, \varepsilon_{it}\} \in A_k \times F_k \subset A \times F \quad (3.12)$$

For now, consider $T_a = \{\lambda_t, \varepsilon_t\} = \{0.935, 0\}$ or the average incarcerated person type. This hypothetical incarcerated person describes the estimated average tastes of the incarcerated people. Using this example, a few welfare implications are drawn below.

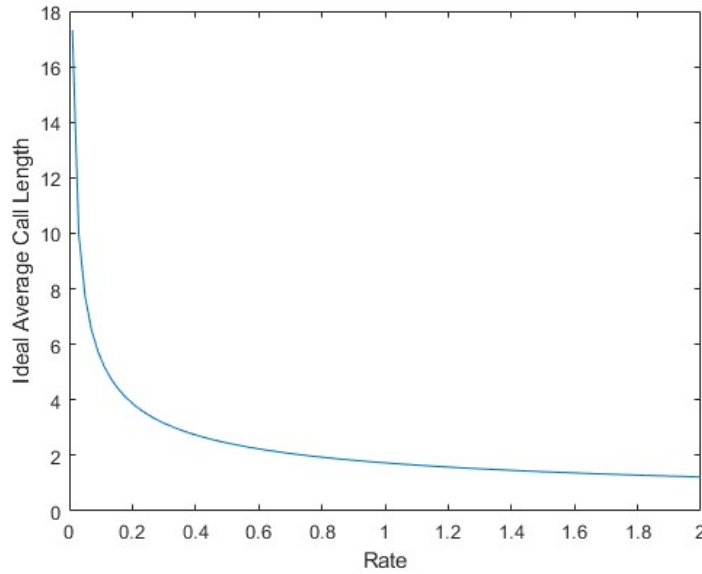


Fig. 3.2 *Optimal Call Length with Respect to Each Per-Min Rate for the Average Incarcerated Person* Optimal call length decreases exponentially with the per-min rate of a call.

Figure 3.2 describes the optimal call length (response) to each possible per-min rate for this average incarcerated person. Note that once the per-min rate is below \$0.2, the optimal call length begins to quickly increase. In the CTC data, the maximum per-min rate is \$0.41, which corresponds to an optimal call length of 2.28 minutes.

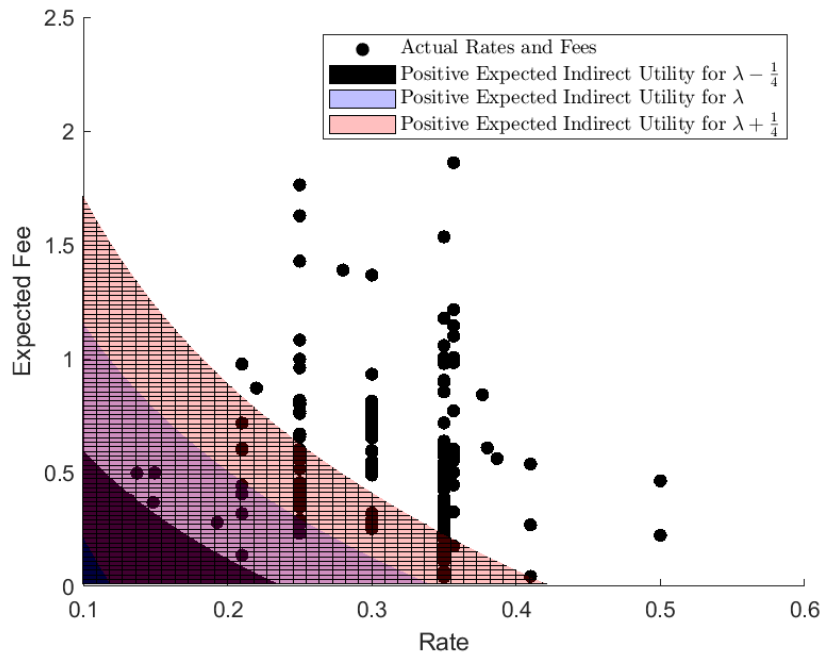


Fig. 3.3 *Positive Region for Expected Indirect Utility with Respect to Per-Min Rates and Fees for the Average incarcerated Person against Actual Rate and Expected Fees* Note that very few of the actual rate and fee observations allow for positive expected indirect utility to be drawn by the average incarcerated person.

In Figure 3.3, the purple (middle) shaded region describes the per-min rate and fees that allow for positive utility to be drawn by the average incarcerated person ($\beta_{it} = \lambda_t = 0.935$). The black (smaller) region describes the same for an incarcerated person with a bit of a below-average taste shock ($\beta_{it} = 0.685$), and the larger (salmon) region describes the same for an incarcerated person with a bit of an above-average taste shock ($\beta_{it} = 1.185$). This motivates the idea that the incarcerated person's (consumer) welfare is quite limited under the current scheme. This discussion of welfare is not complete though. I now move to explore a variety of counterfactuals, each with its own welfare implications.

3.4 Counterfactuals

To address some more general implications of our analysis up to this point, I explore the demand function under 6 policy settings: two with rate caps based on each jail's average daily population (with and without fees), two with uniform rate caps (with and without fees), and two with no rate cap (with and without fees). The first setting is largely based on recent FCC policy [3].

To empirically describe each setting I use the estimates that $\beta_{it} \sim \exp(0.935)$ and $\varepsilon_{it} \sim \mathcal{N}(0, 1)$ to draw 1,000 different prisoner types T_k for each counterfactual m , where each T^m is a sampling of 1,000 T_k from $A \times F$. Each drawn prisoner type is also randomly assigned to a jail, which means that they experience that jail's rate (which could be capped) and expected fee (which could be zero under a ban) under the given scenario.

I then calculate the utility for each prisoner under their counterfactual scheme and their optimal call length. From this generated data I then find the average utility, average call length, and the propensity to call under each counterfactual. Before examining the results of this simulation though, see that each counterfactual scheme is described in further detail below.

3.4.1 ADP-Based Rate Caps and Fee Bans

In July of 2024, the FCC imposed rate caps on contracts the IPCS market based on the relevant jails average daily population according to the schedule below in its *2024 Report & Order* [3].

ADP Range	Rate Cap
1,000+	\$0.06
350-999	\$0.07
100-349	\$0.09
0-99	\$0.12

Table 3.1 *ADP-Based Audio IPCS Per-Minute Rate Caps for U.S. Jails [3]*

With these rate caps the FCC also imposed a ban on ancillary service fees as they state that their imposed rate caps incorporate the "recovery of the costs of ancillary services" [3]. I examine both a scenario with such a fee ban and one without.

3.4.2 Uniform Rate Caps and Fee Bans

As an alternative to the FCC policy, I consider a policy setting with uniform rate caps, where all U.S. jails are subject to the same rate caps regardless of average daily population. The rate cap itself is based on the average rate cap experience under the FCC policy, \$0.11. Here too, I explore a scenario with a fee ban and one without a fee ban.

3.4.3 No Rate Caps

These final two scenarios are closest to the climate studied in the previous sections, as there was no rate cap enforced. However, I do also consider the scenario where there is a fee ban, just without a rate cap to go along with it.

3.4.4 Simulation Results

The simulation as described above provided the following results in Table 3.2. The results from the 6 counterfactual simulations contained a number of metrics. $\mathbb{E}[l_{it}|\cdot]$ is the mean

Rate Cap	Fee Ban	$\mathbb{E}[l_{it} \cdot]$	$\mathbb{E}[u_{it} \cdot]$	$\mathbb{E}[R_{it} \cdot]$	$\mathbb{P}(l_{it} > 0 \cdot)$
None	No	3.1192	0.4111	1.3667	0.2760
None	Yes	2.9229	0.5642	0.8853	0.3930
Uniform	No	8.4668	1.1258	1.3517	0.4810
Uniform	Yes	8.5774	1.4790	0.9229	0.7260
ADP-Based	No	9.0514	1.2149	1.3877	0.4830
ADP-Based	Yes	9.3310	1.5927	0.9514	0.7320

Table 3.2 *Counterfactual Simulation Results*. Note that R_{it} (revenue) is the total between the state and CTC.

call length given the set of $\{\beta_{it}, \varepsilon_{it}\}$ along with the rates r_{it} and fees $\mathbf{F}_{it}$. $\mathbb{E}[u_{it}|\cdot]$ is the mean indirect utility derived by the prisoners given the same information. $\mathbb{E}[R_{it}|\cdot]$ is the mean revenue earned by the state and CTC from the phone calls given l_{it} , r_{it} , and $\mathbf{F}_{it}$. Lastly, $\mathbb{P}(l_{it} > 0|\cdot)$ is the propensity of an arbitrary prisoner to make a call given r_{it} , and $\mathbf{F}_{it}$ along with the set of $\{\beta_{it}, \varepsilon_{it}\}$. These results are discussed in the next section.

Chapter 4

Discussion

The most immediate implication to draw from Table 3.2 is that both rate caps surely improve consumer welfare. This appears to be due to the fact that the rate caps increase the average call lengths and the propensity to make a call. In fact, the average call length nearly tripled with either rate cap scheme. Note that when a fee ban is implemented without a rate cap, the average call length experiences little change. In this case, the increase in consumer welfare appears to be almost solely through the fact that more consumers chose to pick up the phone (up to 39.3 percent from 27.6 percent). Now, it should also be noted that the fee bans appear to have only a relatively small (but noticeable) impact on average utility, while both rate cap schemes nearly triple average utility by themselves.

Conversely, the fee bans do seem to make a substantial impact on the state's and CTC's average revenue. Do note that according to the FCC, the marginal cost of providing a call is approximately negligible for an IPCS provider, and as such average revenue is a good close proxy for average profit [3]. An extended discussion of revenue is beyond the scope of this paper. The revenue-maximizing problem for the state and IPCS provider deeply involves the auction mechanism where per-min rates and commissions (which are not explored in this paper at all) are set, and that mechanism is only considered insofar as its outcomes are taken

by consumers (prisoners). That all being said, it is interesting that revenue is not substantially impacted by the rate caps themselves. It appears that the increase in the number of calls more or less makes up for the capped per-min rates.

Comparing the policies on the basis of total welfare gives the prize to either rate cap scheme without a fee ban. However, still having a propensity to make a call under 0.50 does not seem ideal. A notion of a 'good' propensity to make a call isn't rigorously defined here, but having one above 0.50 seems ideal. Implementing a fee ban on top of one of the rate cap schemes does push the propensity to make a call above this threshold. All of these metrics are somewhat difficult to compare in totality but would surely be useful to consider when designing an auction mechanism that maximizes consumer, total welfare, or some other item on a policy agenda.

The impact on these policies can also be considered with respect to certain demographics, utilizing the results of the OLS models in Chapter 3 and a quick correlation test. First, recall from Chapter 3, that women and those in private jails tended to experience higher than average per-min intrastate rates (though weakly) and that women and U.S. citizens tended to experience a higher than average expected cost from fees (again weakly). So, those in private jails and women may benefit more than others from a rate cap scheme and women and U.S. citizens may benefit more than others from a fee ban. Secondly, a quick correlation test reveals that the ADP of the jails in the CTC subset are positively correlated with the urban indicator ($\rho = 0.56$) and with the proportion of Black inmates ($\rho = 0.36$). Note that the proportion of Black inmates and the urban indicator are not well correlated across the jails in the CTC subset. So, the ADP-based rate cap itself may especially benefit Black inmates and those in urban jails.

Whatever implications to be drawn from the results of the counterfactual simulations are interesting. Though of course, these results are incomplete as they do not factor in the full story of the IPCS market by holding the auction mechanism as fixed and disregarding commissions. However, these results provide some useful motivations for a further analysis considering the auction mechanism and counterfactuals that maximize either consumer welfare, total welfare, or some other objective (i.e. improving welfare in urban jails). Such an analysis would certainly be an interesting extension of this paper's findings. Another potentially interesting extension would be to estimate a demand model that allows for prisoners to choose between audio and video IPCS. As of now, video IPCS is not very prevalent within the available data, but it may become available, as the service becomes more commonplace in U.S. jails. Through this analysis, I have described a consumer demand function through structural equation modeling. This is of course, just one story of the IPCS market, but it does provide a basis for further research into welfare in the IPCS market, which as the policy discussion continues in the United States communications circles, could be quite impactful.

References

[DOJ] *Census of Jails, 2019 Codebook*.

[2] Commission, F. C. (2013). Rates for inmate calling services.

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[4] Marra, M., Miller, N. H., and Sileo, G. (2024). Phoning home: The procurement of telecommunications services for prison systems in the united states.

[5] Marx, K. and Engels, F. (2017). Marx & engels: Collected works.

[6] Moyer, J. (2012). After almost a decade, fcc has yet to rule on high cost of prison phone calls. *Washington Post*.

Appendix A

Data Notes

A.1 DOJ Census of Jails

Variable	Description	Type
Overcrowding Indicator	Indicator for jails above max capacity	Binary
Private Indicator	Indicator for private jails	Binary
Urban Indicator	Indicator for jails located in urban area	Binary
Share of Females	Proportion of adult females	Proportion
Share of Juveniles	Proportion of juveniles	Proportion
Share of Confined Inmates	Proportion of confined inmates	Proportion
Share of Blacked Inmates	Proportion of Black or African-American inmates	Proportion
Share of Hispanic Inmates	Proportion of Black or Latino inmates	Proportion
Share of Inmates who are U.S. Citizens	Proportion of inmates who are U.S. citizens	Proportion
Share of Convicted Inmates	Proportion of inmates who have been convicted	Proportion
Share of Felons	Proportion of inmates who have been convicted of a felony and are serving more than 1 year in jail	Proportion
Share of Inmates under Federal Holdings	Proportion of inmates who are being held at the request of a federal agency	Proportion

Table A.1 *Variable Descriptions for Selected Jail-Level Variables in the 2019 DOJ Census of Jails [DOJ]*

A.2 FCC Form 2301 and FCC Form 2303 for CTC in 2022

Variable	Description	Type
ID	Unique number assigned by provider	Alphanumeric
Total Intrastate Minutes Serviced	Total audio communication minutes the provider serviced	Discrete
Total Intrastate Calls	Total audio communications the provider serviced	Discrete
Weighted Intrastate 15-min Rate	Rate for a 15-minute audio communication calculated as described in Chapter 1	Continuous
Auto Pay Fee Frequency	Auto pay charges divided by total audio communications the provider serviced	Continuous
Billing Fee Frequency	Paper billing charges divided by total audio communications the provider serviced	Continuous
Live Agent Fee Frequency	Live agent charges divided by total audio communications the provider serviced	Continuous
3rd Party Transaction Fee Frequency	3rd party transaction charges divided by total audio communications the provider serviced	Continuous
ADP	Average daily population	Discrete

Table A.2 *Variable Descriptions for Selected Jail-Level Variables from CTC's 2022 submission of FCC Form 2301 [3]*

Appendix B

Additional Counterfactual Notes

B.1 Histograms

Below are the complete results of each counterfactual simulation. Each observation is the result of a potential phone call. Recall that if a phone call is not made (the outside option), the prisoner receives no utility.

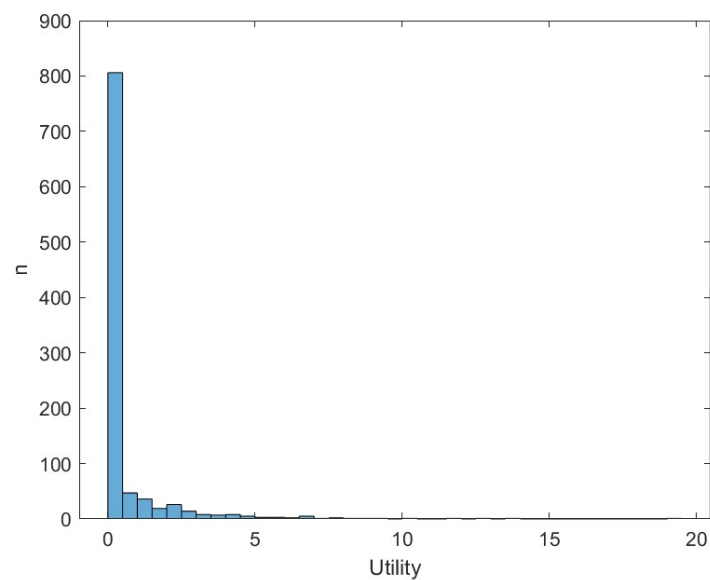


Fig. B.1 *Counterfactual Results for No Rate Caps with Fees*

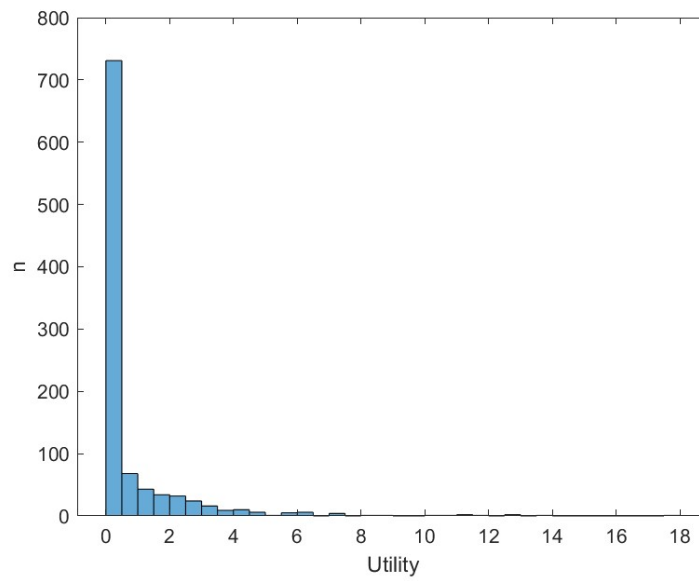


Fig. B.2 Counterfactual Simulation Results for No Rate Caps and No Fees

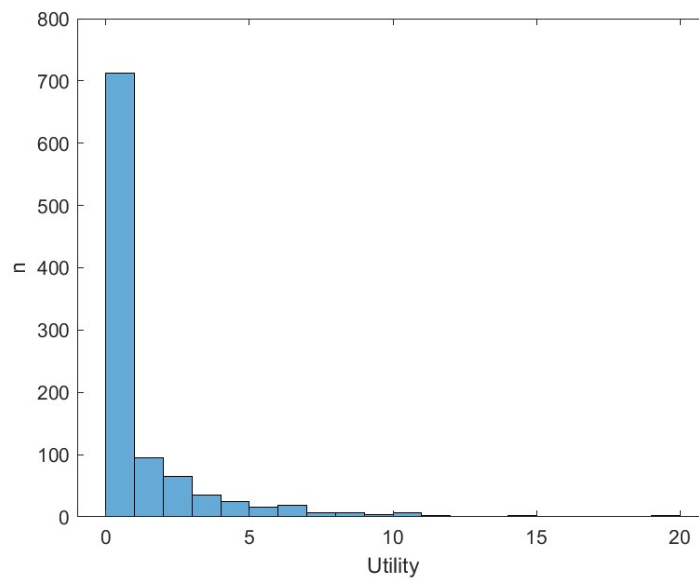


Fig. B.3 Counterfactual Simulation Results for Uniform Rate Caps with Fees

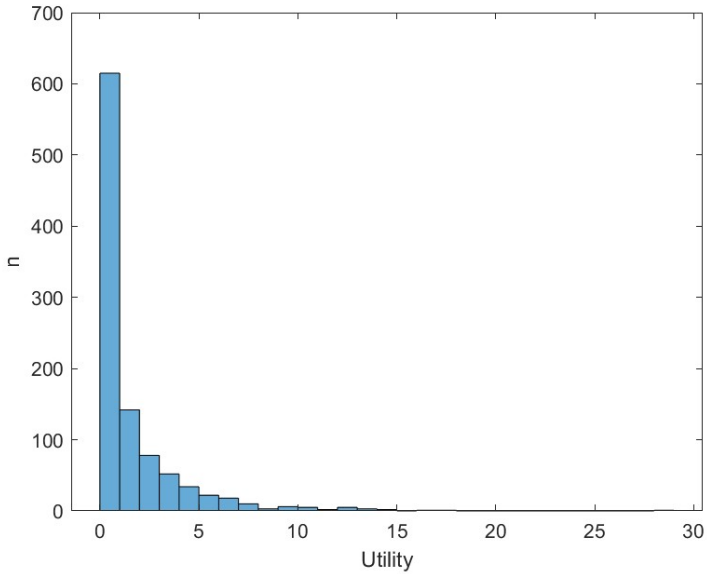


Fig. B.4 Counterfactual Simulation Results for Uniform Caps with No Fees

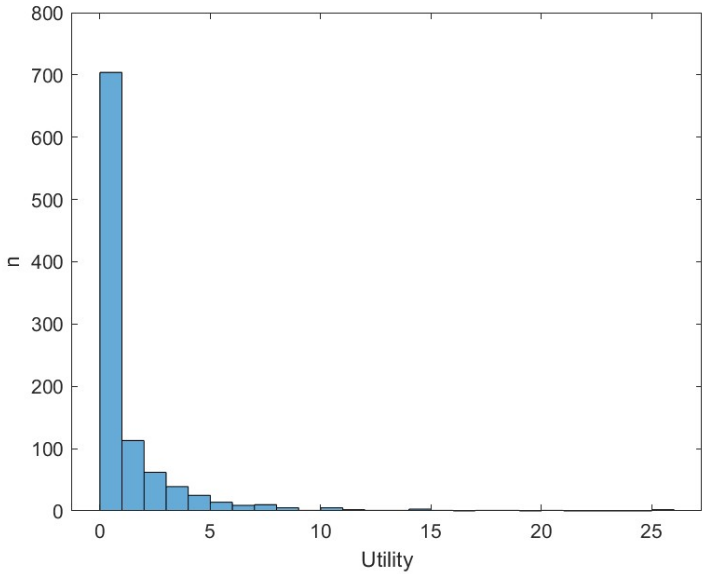


Fig. B.5 Counterfactual Simulation Results for ADP-Based Rate Caps with Fees

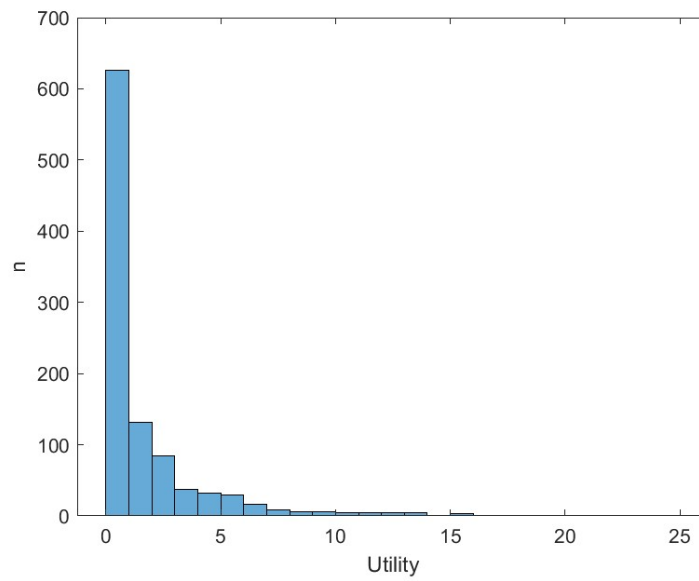


Fig. B.6 Counterfactual Simulation Results for ADP-Based Rate Caps with No Fees

B.2 MATLAB Code

Below is the MATLAB code used to perform the counterfactual simulation as described in Chapter 5.

```
% clear workspace and console
clear;
clc;

% rates and fee data for different counterfactuals
CTCRates = readtable('CTC_Rates_Fee_Costs2.csv');
r = CTCRates.weighted_Intra_15_Avg/15;
r_fcc = CTCRates.Intra_Rate_FCC/15;
r_unif = CTCRates.Intra_Rate_Unif/15;
fee = CTCRates.Total_Expected_Fee_Cost;
transac = CTCRates.Transac_Cost;

% estimated pars
Lam = 0.935;
Mu = 0;
Sigma = 1;

% empty results (call length, util, rev, prop) matrix
results = zeros(6,4);

% sim data for base
beta_1 = exprnd(Lam,[1000,1]);
epsilon_1 = normrnd(Mu,Sigma,[1000,1]);
```

```
jail_1 = randi([1 188],[1000,1]);

sample_1 = zeros(1000, 3);

for i = 1:1000
    sample_1(i,1) = r(jail_1(i));
    sample_1(i,2) = fee(jail_1(i));
    sample_1(i,3) = transac(jail_1(i));
end

data_1 = [beta_1, epsilon_1, sample_1(:,1), sample_1(:,2), sample_1(:,3)];

% calc results
lens_1 = beta_1./(sample_1(:,1));
utils_1 = beta_1.*log(lens_1) - (lens_1).*(sample_1(:,1))-sample_1(:,2);
utils_1(utils_1 < 0) = 0;
revs_1 = (lens_1).*(sample_1(:,1))+sample_1(:,2)-sample_1(:,3);

results(1,1) = mean(lens_1);
results(1,2) = mean(utils_1);
results(1,3) = mean(revs_1);
results(1,4) = (numel(utils_1)-sum(utils_1 == 0))/numel(utils_1);

% sim data for base - fees
beta_2 = exprnd(Lam,[1000,1]);
epsilon_2 = normrnd(Mu,Sigma,[1000,1]);
```

```
jail_2 = randi([1 188],[1000,1]);

sample_2 = zeros(1000, 3);

for i = 1:1000
    sample_2(i,1) = r(jail_2(i));
    sample_2(i,2) = fee(jail_2(i));
    sample_2(i,3) = transac(jail_2(i));
end

data_2 = [beta_2, epsilon_2, sample_2(:,1), sample_2(:,2), sample_2(:,3)];

% calc results
lens_2 = beta_2./(sample_2(:,1));
utils_2 = beta_2.*log(lens_2) - (lens_2).*(sample_2(:,1));
utils_2(utils_2 < 0) = 0;
revs_2 = (lens_2).*(sample_2(:,1));

results(2,1) = mean(lens_2);
results(2,2) = mean(utils_2);
results(2,3) = mean(revs_2);
results(2,4) = (numel(utils_2)-sum(utils_2 == 0))/numel(utils_2);

% sim data for uni cap
beta_3 = exprnd(Lam,[1000,1]);
epsilon_3 = normrnd(Mu,Sigma,[1000,1]);
```

```
jail_3 = randi([1 188],[1000,1]);

sample_3 = zeros(1000, 3);

for i = 1:1000
    sample_3(i,1) = r_unif(jail_3(i));
    sample_3(i,2) = fee(jail_3(i));
    sample_3(i,3) = transac(jail_3(i));
end

data_3 = [beta_3, epsilon_3, sample_3(:,1), sample_3(:,2), sample_3(:,3)];

% calc results
lens_3 = beta_3./(sample_3(:,1));
utils_3 = beta_3.*log(lens_3) - (lens_3).*(sample_3(:,1))-sample_3(:,2);
utils_3(utils_3 < 0) = 0;
revs_3 = (lens_3).*(sample_3(:,1))+sample_3(:,2)-sample_3(:,3);

results(3,1) = mean(lens_3);
results(3,2) = mean(utils_3);
results(3,3) = mean(revs_3);
results(3,4) = (numel(utils_3)-sum(utils_3 == 0))/numel(utils_3);

% sim data for uni cap - fees
beta_4 = exprnd(Lam,[1000,1]);
epsilon_4 = normrnd(Mu,Sigma,[1000,1]);
```

```
jail_4 = randi([1 188],[1000,1]);

sample_4 = zeros(1000, 3);

for i = 1:1000
    sample_4(i,1) = r_unif(jail_4(i));
    sample_4(i,2) = fee(jail_4(i));
    sample_4(i,3) = transac(jail_4(i));
end

data_4 = [beta_4, epsilon_4, sample_4(:,1), sample_4(:,2), sample_4(:,3)];

% calc results
lens_4 = beta_4./(sample_4(:,1));
utils_4 = beta_4.*log(lens_4) - (lens_4).*(sample_4(:,1));
utils_4(utils_4 < 0) = 0;
revs_4 = (lens_4).*(sample_4(:,1));

results(4,1) = mean(lens_4);
results(4,2) = mean(utils_4);
results(4,3) = mean(revs_4);
results(4,4) = (numel(utils_4)-sum(utils_4 == 0))/numel(utils_4);

% sim data for fcc cap
beta_5 = exprnd(Lam,[1000,1]);
epsilon_5 = normrnd(Mu,Sigma,[1000,1]);
```

```
jail_5 = randi([1 188],[1000,1]);

sample_5 = zeros(1000, 3);

for i = 1:1000
    sample_5(i,1) = r_fcc(jail_5(i));
    sample_5(i,2) = fee(jail_5(i));
    sample_5(i,3) = transac(jail_5(i));
end

data_5 = [beta_5, epsilon_5, sample_5(:,1), sample_5(:,2), sample_5(:,3)];

% calc results
lens_5 = beta_5./(sample_5(:,1));
utils_5 = beta_5.*log(lens_5) - (lens_5).*(sample_5(:,1))-sample_5(:,2);
utils_5(utils_5 < 0) = 0;
revs_5 = (lens_5).*(sample_5(:,1))+sample_5(:,2)-sample_5(:,3);

results(5,1) = mean(lens_5);
results(5,2) = mean(utils_5);
results(5,3) = mean(revs_5);
results(5,4) = (numel(utils_5)-sum(utils_5 == 0))/numel(utils_5);

% sim data for uni cap - fees
beta_6 = exprnd(Lam,[1000,1]);
epsilon_6 = normrnd(Mu,Sigma,[1000,1]);
```

```
jail_6 = randi([1 188],[1000,1]);

sample_6 = zeros(1000, 3);

for i = 1:1000
    sample_6(i,1) = r_fcc(jail_6(i));
    sample_6(i,2) = fee(jail_6(i));
    sample_6(i,3) = transac(jail_6(i));
end

data_6 = [beta_6, epsilon_6, sample_6(:,1), sample_6(:,2), sample_6(:,3)];
% calc results
lens_6 = beta_6./(sample_6(:,1));
utils_6 = beta_6.*log(lens_6) - (lens_6).*(sample_6(:,1));
utils_6(utils_6 < 0) = 0;
revs_6 = (lens_6).*(sample_6(:,1));

results(6,1) = mean(lens_6);
results(6,2) = mean(utils_6);
results(6,3) = mean(revs_6);
results(6,4) = (numel(utils_6)-sum(utils_6 == 0))/numel(utils_6);

% histograms
figure
histogram(utils_1);
xlabel('Utility')
```

```
ylabel('n')
```

```
figure  
histogram(utills_2);  
xlabel('Utility')  
ylabel('n')
```

```
figure  
histogram(utills_3);  
xlabel('Utility')  
ylabel('n')
```

```
figure  
histogram(utills_4);  
xlabel('Utility')  
ylabel('n')
```

```
figure  
histogram(utills_5);  
xlabel('Utility')  
ylabel('n')
```

```
figure  
histogram(utills_6);  
xlabel('Utility')  
ylabel('n')
```

Appendix C

On Marx's Quote

I will admit the quote at the beginning of this paper is a bit odd. Hemorrhoids and the French Bourgeois are not typically discussed together, much less in the economics literature. Thus, I will take a moment to explain my motivation for including it in this paper.

I first came across the quote when researching the origins of the European Monetary Union. I was in high school at the time, writing my capstone paper on 'memories' of German Reunification, i.e. what West and East German citizens personally accounted about the period following the reunification. What I found striking was that while many American historians lauded the reunification as a historic moment for German national pride and the battle of capitalistic democracy over communist authoritarianism, German citizens themselves far more often noted concerns about inflation and market consolidation when describing the reunification. They were worried about how their trips to the grocery store, visits to the doctor, and vacations would change, not national pride. I think this sentiment echoes Marx. Ultimately, even during historical events, we are most concerned with the little things that affect our day-to-day lives, which, in other words, are economic factors (the cost of groceries, health insurance coverage, or the availability of airplane tickets). It was writing this paper, and ipso facto reading Marx's quote, in high school that made me want to study economics

(albeit economic history), and so I include it here in this paper as a reminder to myself what this research effort is all about– the little things.